

**VIT[®]****Vellore Institute of Technology**
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Vellore – 632014, Tamil Nadu, India

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**Win Semester 2025-2026****Continuous Assessment-2****Patent Technical Content**

SLOT : D1+TD1
Programme Name : B.Tech.
Course Name & code : Introduction to Internet of Things & CSE1030
Class Number : VL2025260505493
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Team No : 19

Page 2: Technical Description & Patent Preparation

Part E: System Architecture & Working Principle

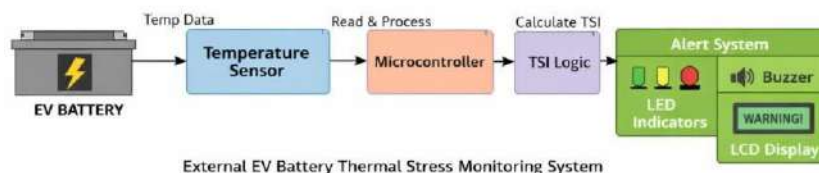
System Architecture

The system consists of an external temperature sensor, a microcontroller, and alert units such as LEDs, buzzer, and an LCD display. The sensor is attached to the outer surface of the EV battery and works independently of the internal BMS.

Working Principle

The temperature sensor continuously measures the battery surface temperature. The microcontroller analyses the temperature trend and calculates the Thermal Stress Index (TSI) based on the rate of temperature rise. According to the TSI level, the system provides visual and audio alerts to the driver to indicate safe, warning, or danger conditions.

Block Diagram:



Components Used:

1. Temperature Sensor
 - Measures battery surface temperature continuously.
2. MicroPython On Raspberry Pi
 - Reads sensor data and processes it.
3. Alert System (LEDs & Buzzer)
 - Provides visual and audio warning to the driver.
4. LCD Display
 - Displays battery status in simple words.

Workflow / Algorithm:

Step 1: The temperature sensor continuously measures the battery surface temperature.

Step 2: The microcontroller records temperature readings at regular time intervals.

Step 3: The system calculates the rate of temperature increase and determines the Thermal Stress Index (TSI).

Step 4: Based on TSI level, the system activates LEDs, buzzer, and displays warning messages to the driver.

Part F: Experimental Validation & Results

How was the system tested?

The system was tested by:

- ❖ Simulating normal heating conditions.
- ❖ Simulating sudden abnormal temperature rise.
- ❖ Observing alert response accuracy.

What were the outcomes?

- ❖ The system successfully detected abnormal battery heating early.
- ❖ LED and buzzer alerts responded correctly to risk levels.
- ❖ The system consumed low power and worked reliably.

Test Results Table:

Test Parameter	Method	Result
Detection Accuracy	Compared predicted risk vs temperature trend	High
False Positive Rate	Tested under normal heat conditions	Low
Power Consumption	Measured during operation	Low
Alert Success Rate	Verified LED & buzzer response	Reliable
Model Performance	Based on early detection capability	Effective

Key Findings:

- The system successfully detected abnormal temperature rise early.
- False alerts due to environmental heat were minimized.
- Alerts were clear and easily understood by drivers.

Part G: Prior Art & Comparative Analysis

Existing Solution	Key Features	How our invention is different
Internal BMS	Fixed temperature cut-off	Our system predicts risk using temperature rise behaviour.
Dashboard Temperature Display	Shows battery temperature value	Our system converts data into simple alerts.
App-Based Monitoring	Requires smartphone & internet	Our system works offline with instant alerts.

Part H: Patentable Aspects

- 🚦 Thermal Stress Index (TSI) based on temperature rise behaviour.
- 🚦 Rate-of-temperature-change based prediction logic.
- 🚦 External, plug-and-play battery safety system independent of BMS.
- 🚦 Driver-friendly alert mechanism using LEDs and buzzer.

Part I: Technology Readiness Level (TRL)

- ✅ TRL 4 – Technology validated in laboratory environment

The system logic and alert mechanism have been tested using simulated temperature conditions and demonstrated successful early warning capability.

IOT Level Used :

➤ Level 2

Device → Cloud → User (Two-way communication)

It includes:

- Sensor data collection.
- Internet connectivity.
- Cloud storage.
- User dashboard.
- Alerts to user.

Why Level-2 Fits Your EV Battery Prediction Model?

- DS18B20 Temperature Sensor.
- Raspberry Pi MicroPython (WiFi enabled).
- TSI calculation (local processing).
- Cloud monitoring (ThingSpeak / MQTT).
- User alert (LED / buzzer + mobile dashboard).

Implementation of the model using raspberry Pi MicroPython:

WOKWI

main.py diagram.json

```
1 from machine import Pin
2 import onewire
3 import ds18x20
4 import time
5
6 # DS18B20 Setup
7 datapin = Pin(16) # GP16
8 ow = onewire.OneWire(datapin)
9 ds = ds18x20.DS18x20(ow)
10 roms = ds.scan()
11
12 # LEDs
13 green = Pin(15, Pin.OUT)
14 yellow = Pin(14, Pin.OUT)
15 red = Pin(13, Pin.OUT)
16
17 # Buzzer
18 buzzer = Pin(12, Pin.OUT)
19
20 # Constants
21 Tsafe = 60.0
22 delta_t = 2.0 # minutes
23
24 # Initial temperature
25 ds.convert_temp()
26 time.sleep(1)
27 T_prev = ds.read_temp(roms[0])
28
29 while True:
30     time.sleep(10) # 2 minutes delay
31
32     ds.convert_temp()
33     time.sleep(1)
34     T_curr = ds.read_temp(roms[0])
35
36     # TSI calculation
37     TSI = (T_curr - T_prev) / (delta_t * Tsafe)
38
39     print("Current Temp:", T_curr)
40     print("TSI:", TSI)
```

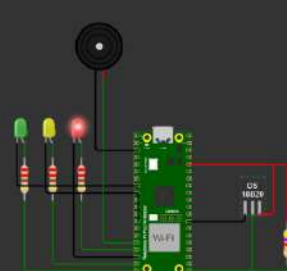
Simulation

01:01:230 100%

DALLAS DS18B20 TEMPERATURE SENSOR

Temperature: 107.2°C

Device Address: 28 01 02 03 04 05 06 (default)



Current Temp: 125.0
TSI: 0.0
Current Temp: 74.0
TSI: -0.425
Current Temp: 93.1875
TSI: 0.1598958
Current Temp: 93.1875
TSI: 0.0
Current Temp: 107.1875
TSI: 0.1166667